

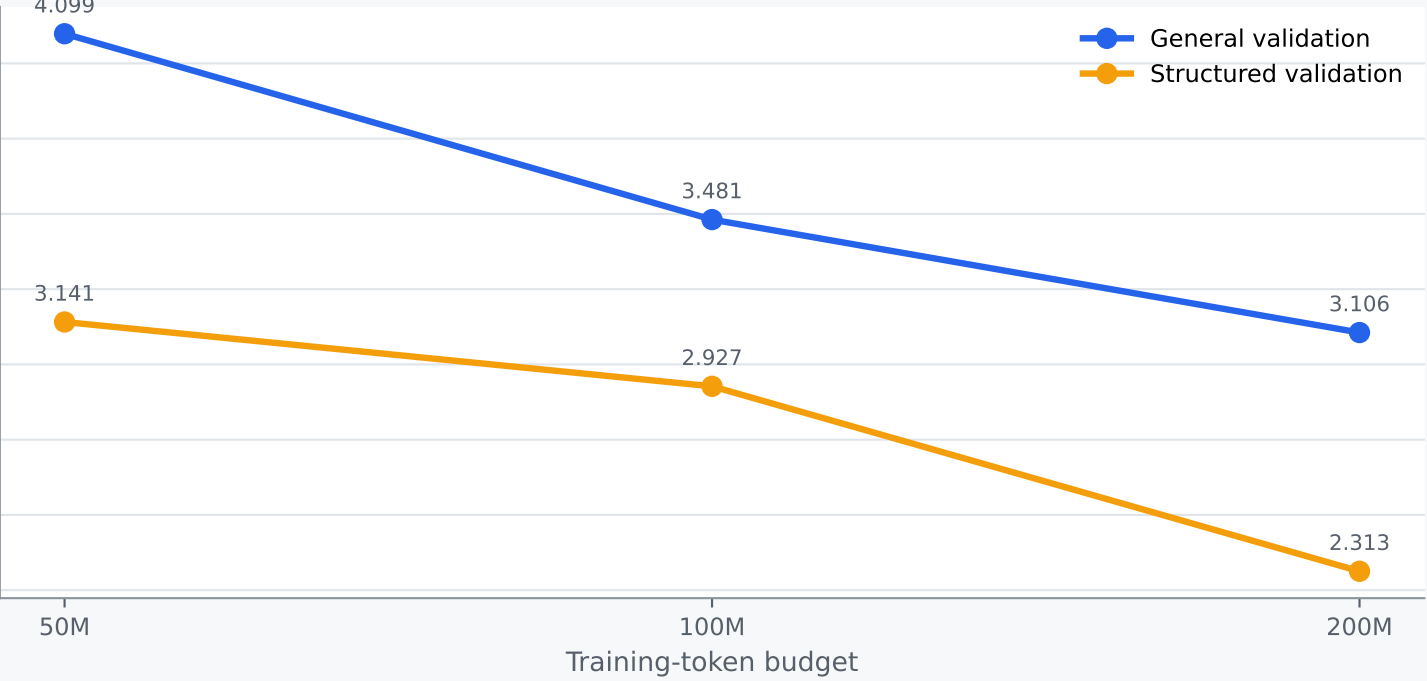
ToolCall M30 scaling study

Fixed 30.0M-parameter model · comparison across training-token budgets

COMPLETED

3/3 runs

Final validation loss vs token budget



BEST GENERAL LOSS

3.1059 · High · 200M

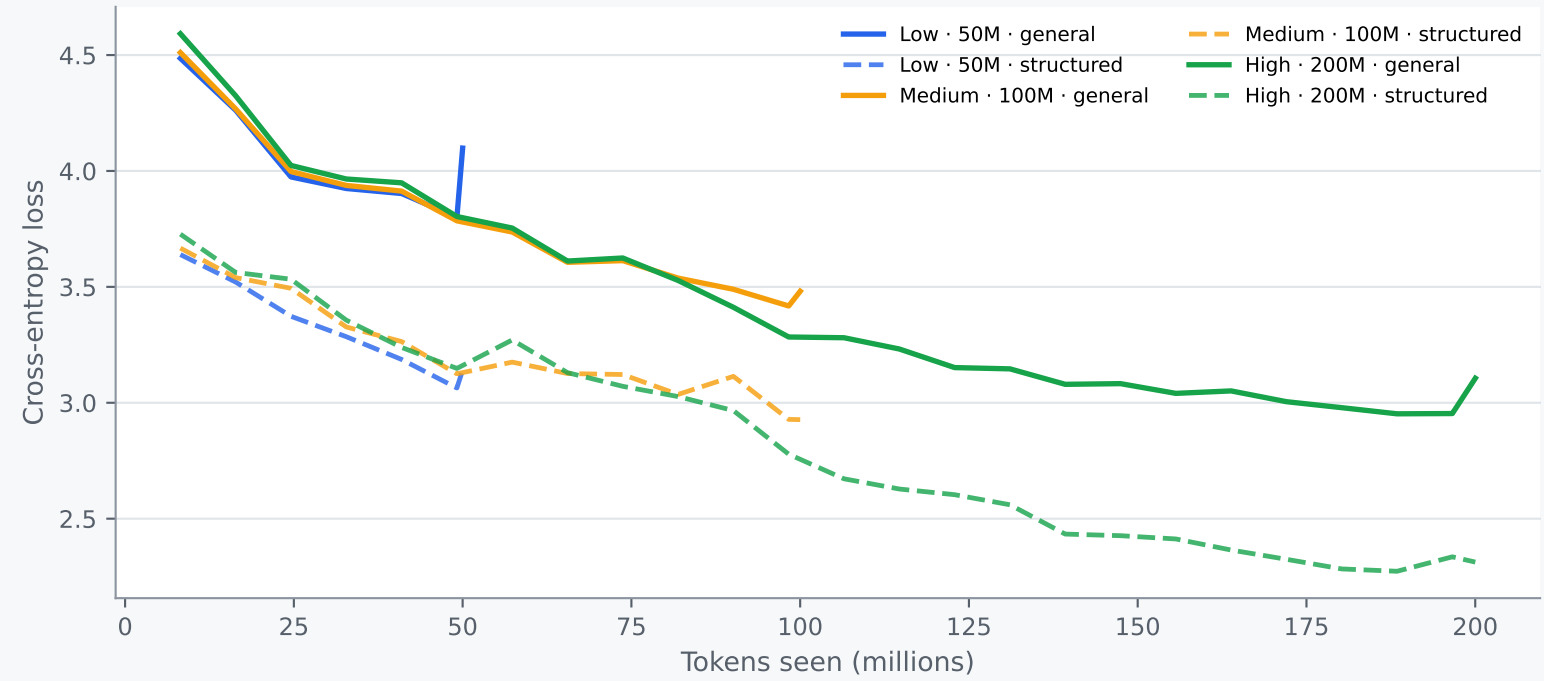
BEST STRUCTURED LOSS

2.3128 · High · 200M

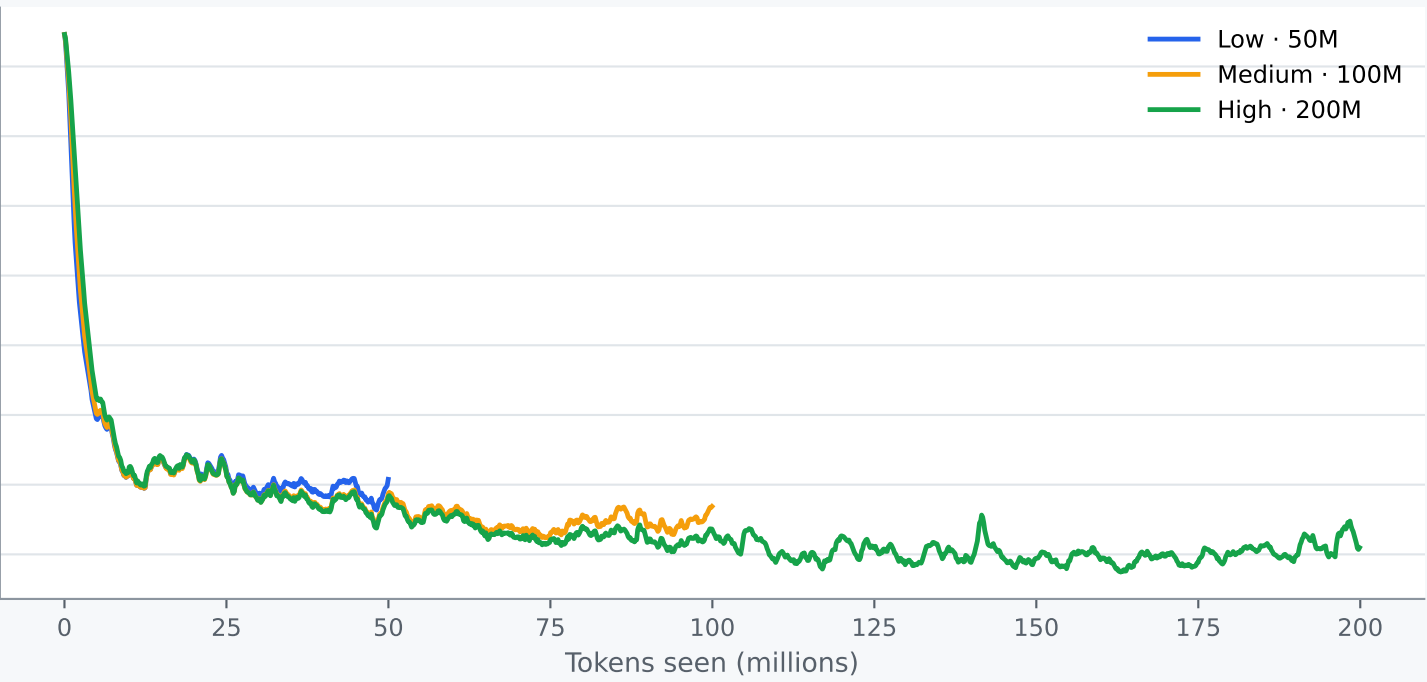
TOTAL GPU TIME

2.54 hours

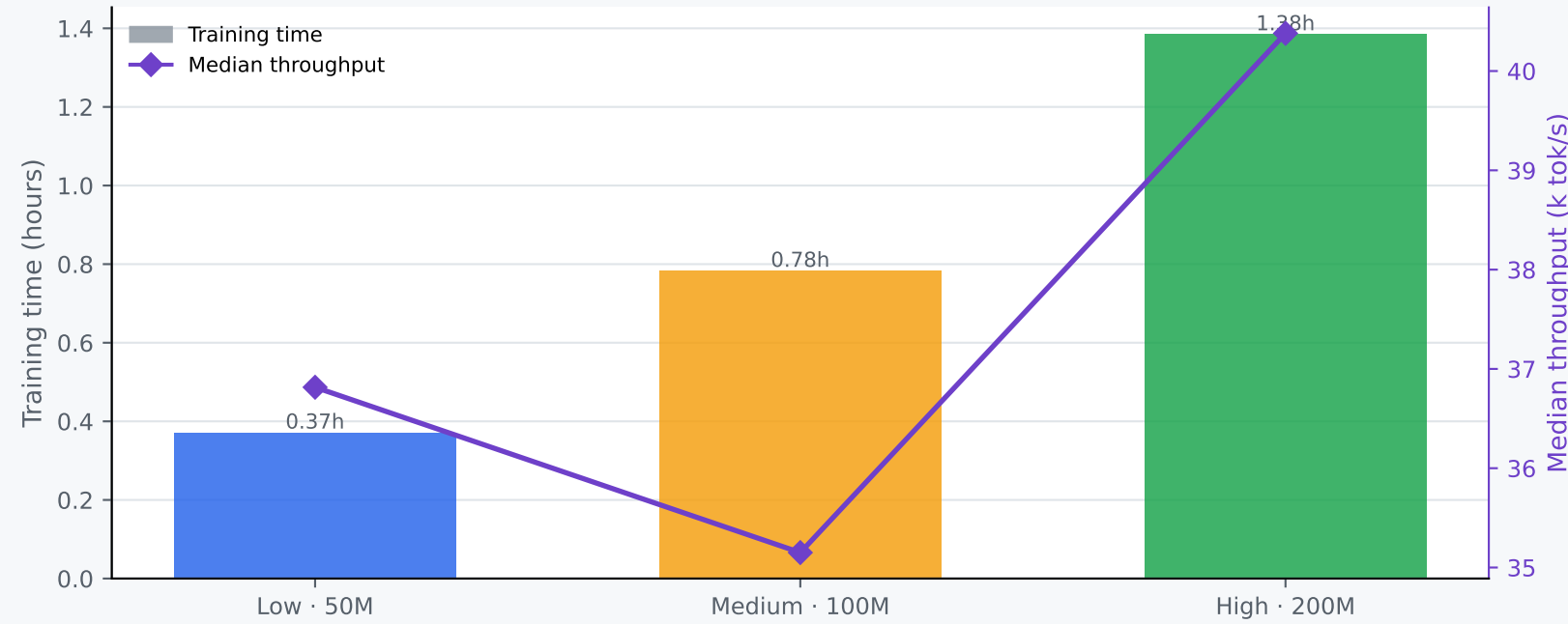
Validation learning curves



Training-loss convergence



Runtime and throughput



Run comparison

Run	Status	Tokens	General loss	Structured loss	Train time	Median speed	Peak VRAM
Low · 50M	completed	50.0M	4.0986	3.1409	0.37 h	36.8k tok/s	2.93 GiB
Medium · 100M	completed	100.0M	3.4811	2.9272	0.78 h	35.2k tok/s	2.93 GiB
High · 200M	completed	200.0M	3.1059	2.3128	1.38 h	40.4k tok/s	2.93 GiB